

EVision Bescom

**Behavior-Aware EV Charging & Infrastructure
Intelligence for BESCOM**

1. Introduction

Electric Vehicle (EV) adoption in Bengaluru is rapidly increasing, leading to new challenges in managing electricity demand. EV charging demand is highly concentrated in specific locations and time periods, especially during evening hours. This creates localized stress on the power distribution network.

EVision AI is designed as an intelligent decision-support system that helps optimize EV charging behavior and infrastructure planning without modifying existing grid systems.

2. Problem Statement

As EV adoption grows:

- Charging demand becomes clustered in time and space
- Grid infrastructure faces localized overload risks
- Existing solutions:
 - Assume uniform user behavior
 - Do not adapt to real-time grid conditions
 - Lack future planning capability

This leads to inefficient grid usage and poor infrastructure decisions.

3. Objectives

The key objectives of EVision AI are:

- Predict EV charging demand using behavior-aware models
- Optimize charging schedules based on grid conditions
- Provide real-time grid status insights
- Enable scenario-based infrastructure planning
- Deliver actionable recommendations via a dashboard

4. Proposed Solution

EVision AI is a decision-support layer that integrates:

- Behavior-aware demand modeling
- Real-time grid monitoring
- Simulation-based planning
- Optimization-driven recommendations

The system helps grid operators and planners make informed decisions to reduce overload risks and improve efficiency.

5. Key Features

5.1 Behavior-Aware Demand Modeling

Users are categorized into:

- Urgent users (charge immediately)
- Flexible users (can delay charging)
- Habit-based users (fixed routines)

This improves the accuracy of demand prediction.

5.2 Real-Time Grid Response Engine

The system classifies grid zones into:

- Normal
- Warning
- Critical

Based on this, it generates:

- Charging deferral suggestions
- Load redistribution strategies
- Off-peak incentives

5.3 Simulation Engine

Supports “what-if” analysis such as:

- Increase in EV adoption
- New charging station deployment
- Changes in user behavior

Outputs include:

- Grid load impact
- Risk of overload
- Infrastructure effectiveness

5.4 Infrastructure Planning

- Identifies high-demand zones
- Considers grid capacity and future demand
- Recommends optimal charging station locations

5.5 Interactive Dashboard

Provides:

- Live grid status
- Demand visualization
- Recommendations
- Simulation results

6. System Architecture

The system consists of four layers:

Frontend

- Built using React.js
- Displays dashboards, charts, and insights

Backend

- Node.js APIs
- Handles data processing and communication

AI & Optimization Layer

- Demand prediction models
- Grid classification logic
- Optimization algorithms

Data Layer

- Synthetic EV usage data
- Grid load data

7. Working Mechanism

1. Data is collected (EV usage, grid load)
2. Demand prediction is performed using behavior-aware models
3. Grid condition is classified (Normal/Warning/Critical)
4. Simulation engine evaluates future scenarios
5. Optimization engine generates recommendations
6. Results are displayed on the dashboard

8. Technology Stack

- Frontend: React.js
- Backend: Node.js
- Visualization: Chart.js
- Data: Synthetic datasets
- Models: Basic ML & simulation-based logic

9. Results & Outputs

The system provides:

- Real-time grid monitoring
- Demand prediction insights
- Charging recommendations
- Infrastructure planning suggestions
- Scenario simulation results

10. Impact

- Reduces peak load stress on the grid
- Improves charging efficiency
- Enables data-driven planning
- Supports scalable EV ecosystem

11. Future Scope

- Integration with real BESCOM grid data
- Advanced AI models (Reinforcement Learning)
- Mobile app for EV users
- Policy and tariff integration

12. Conclusion

EVision AI bridges the gap between demand prediction, user behavior, and infrastructure planning. It provides a scalable and practical solution for managing EV charging in smart cities, ensuring efficient grid utilization and sustainable growth.